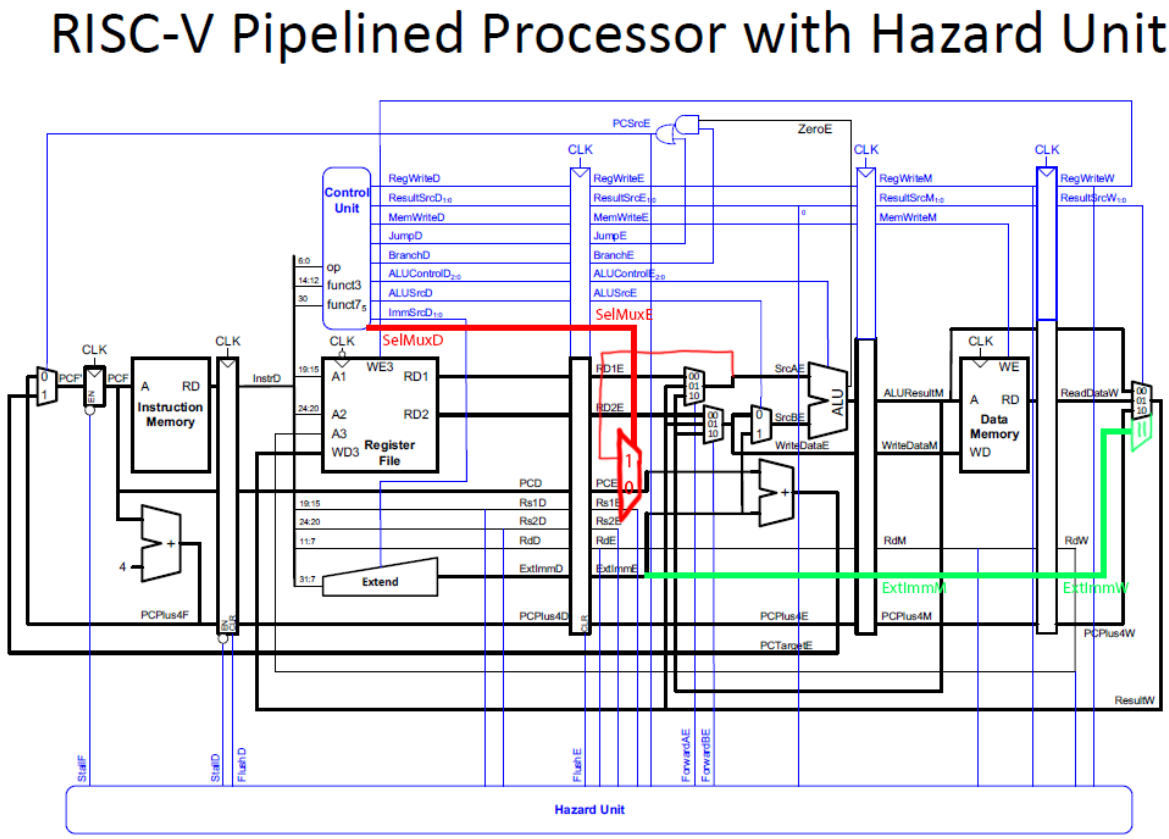
سیستم های دیجیتال 2 | تمرین کامپیوتری 4

متین شیاسی 810101453 | آرین ریاحی 810101433

Datapath:



Controller:

Instruction Type	Instructions	Configurations
R-type	add, sub, and, or, slt	-RegWrite = 1, ALUOp = 10, ALUSrc = 0, MemWrite = 0, ResultSrc = 00, Branch = 0, Jump = 0 ALUFunc: add: 000, sub: 001, and: 010, or: 011, slt: 100
I-type	lw	- RegWrite = 1, ImmSrc = 000, ALUSrc = 1, MemWrite = 0, ResultSrc = 01 - ALUOp = 00, Branch = 0, Jump = 0 - ALUFunc = 000

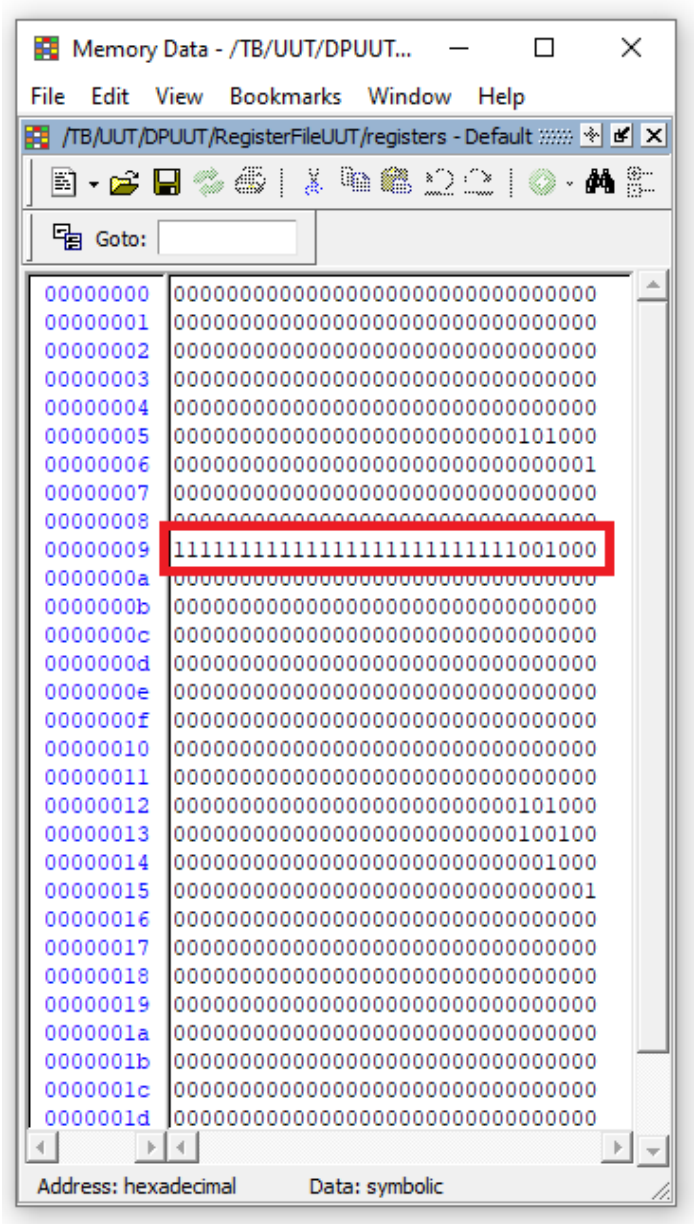
	addi, xori, ori, slti	- RegWrite = 1, ImmSrc = 000, ALUSrc = 1, MemWrite = 0, ResultSrc = 00
		- ALUOp = 11, Branch = 0, Jump = 0
		- ALUFunc:
		  - addi: 000
		  - xori: 100
		  - ori: 011
		  - slti: 100

	jalr	- RegWrite = 1, ImmSrc = 000, ALUSrc = 1, MemWrite = 0, ResultSrc = 10 - ALUOp = 00, SelMux = 1, Branch = 0, Jump = 1 - ALUFunc = 000 - Operation: - jalr: ra = PC + 4, PC = rs1 + adr
S-type	sw	- RegWrite = 0, ImmSrc = 001, ALUSrc = 1, MemWrite = 1, ResultSrc = 00 - ALUOp = 00, Branch = 0, Jump = 0 - ALUFunc = 000
J-type	jal	- RegWrite = 1, ImmSrc = 010, MemWrite = 0, ResultSrc = 10 - ALUOp = 00, SelMux = 1 - Operation: - jal: ra = PC + 4, PC = PC + adr
B-type	beq, bne	- RegWrite = 0, ImmSrc = 001, ALUSrc = 0, MemWrite = 0, ALUOp = 01, ResultSrc = 00 - SelMux = 0, Branch = 1, Jump = 0
U-type	lui	- RegWrite = 1, ImmSrc = 100, MemWrite = 0, ResultSrc = 11 - Branch = 0, Jump = 0

Instructions and Memory:

```
addi x8,x8 , 0x000; ->          00000000000001000000010000010011
addi x5, x0, 1; ->              00000000000010000000001010010011
sw x5, 0(x8); ->                00000000101010000010000000100011
addi x5, x0, 2; ->              00000000000010000000001010010011
sw x5, 4(x8); ->                00000000001010100001000100100011
addi x5, x0, 42; ->             00000010101000000000001010010011
sw x5, 8(x8); ->                00000000001010100001001000100011
addi x5, x0, 32; ->             00000010000000000000001010010011
sw x5, 12(x8); ->               00000000001010100001001100100011
addi x5, x0, 12; ->             00000000110000000000001010010011
sw x5, 16(x8); ->               00000000001010100001010000100011
sw x5, 20(x8); ->               00000000001010100001010100100011
addi x5, x0, -56; ->            11111100100000000000001010010011
sw x5, 24(x8); ->               00000000001010100001011000100011
addi x5, x0, 7; ->              00000000011100000000001010010011
sw x5,28(x8); ->                00000000001010100001011100100011
addi x5, x0, -45; ->            11111101001100000000001010010011
sw x5, 32(x8); ->               00000010010101000001000000100011
addi x5, x0, 8; ->              00000000100000000000001010010011
sw x5, 36(x8); ->               00000010010101000001000100100011
addi x6, x0, 1; ->              00000000000010000000001100010011
lw x9, 0(x8); ->                00000000000000100000100100001011
addi x18, x18, 0; ->            00000000000001001000001001010011
addi x5, x5, 40; ->             00000101000001010000001010010011
loop:
beq x18, x5, end_loop; ->       00000010010110010000000001100011
add x19, x18, x8; ->            00000001001001000000100110011
lw x20, 0(x19); ->              00000000000001001101001010000011
slt x21, x9, x20; ->            00000001010001001010010110011
beq x21, x6, end_if; ->         00000000011010101000000001100011
addi x9, x20, 0; ->             00000000000001010000001001010011
end_if:
addi x18, x18, 4; ->            00000000010010010000010010010011
jal loop; ->                    11111110010111111111111111101111
end_loop:
```

Result:


Memory Data - /TB/UUT/DPUUT...
File Edit View Bookmarks Window Help
/TB/UUT/DPUUT/RegisterFileUUT/registers - Default
Goto:
00000000 00000000000000000000000000000000
00000001 00000000000000000000000000000000
00000002 00000000000000000000000000000000
00000003 00000000000000000000000000000000
00000004 00000000000000000000000000000000
00000005 00000000000000000000000000101000
00000006 00000000000000000000000000000001
00000007 00000000000000000000000000000000
00000009 11111111111111111111111111001000
0000000b 00000000000000000000000000000000
0000000c 00000000000000000000000000000000
0000000d 00000000000000000000000000000000
0000000e 00000000000000000000000000000000
0000000f 00000000000000000000000000000000
00000010 00000000000000000000000000000000
00000011 00000000000000000000000000000000
00000012 00000000000000000000000000101000
00000013 00000000000000000000000000100100
00000014 00000000000000000000000000001000
00000015 00000000000000000000000000000001
00000016 00000000000000000000000000000000
00000017 00000000000000000000000000000000
00000018 00000000000000000000000000000000
00000019 00000000000000000000000000000000
0000001a 00000000000000000000000000000000
0000001b 00000000000000000000000000000000
0000001c 00000000000000000000000000000000
0000001d 00000000000000000000000000000000
Address: hexadecimal
Data: symbolic